A Multidimensional Divide-and-Conquer Algorithm for Assigning Secondary Structures in Proteins

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Abstract

In this paper, we propose a simple algorithm which can automatically assign secondary structures in a protein using a list of nitrogen (N), carbon (C) and oxygen (O) coordinates on its backbone, which can be modeled as sparse points in three dimensional space. Our algorithm has two stages. In the first stage, it determines hydrogen bonds based on pair-wise distances between N and O atoms of difference residues. Then a fragment of consecutive residues is considered as a candidate of α-helix/β-sheet if it follows a specific pattern on the linking of hydrogen bonds. Moreover, if there are at least four CO vectors that are parallel in this fragment, then this fragment is an α-helix; otherwise, it is considered as a β-sheet. A brute-force implementation of the above algorithm requires $O\left(n^2\right)$ time, where n is the number of residues. With the help of multidimensional divide-and-conquer approach commonly used in computational geometry, we reduce the time complexity to $O\left(n \log^2 n + m\right)$, where m is the number of identified hydrogen bonds. Our experimental results show that our results are comparable with those obtained by the DSSP, which is currently the most widely used tool.

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1 Introduction

The problem of assigning secondary structures in proteins is very important, because it has many applications in biology, such as the disulfide bonds prediction of cystenines [7], protein secondary structure prediction [1, 13, 27], search for structurally similar proteins [30], and protein structure alignment [14, 22].

The ordinary procedure for assigning secondary structures in a protein is as follows. First of all, the coordinates of atoms of a protein are determined by X-ray crystallography. Then α-helix, β-sheet, and turn structures are manually annotated by biologists. For small protein structures, this manual assignment is not difficult. However, it becomes a time-consuming job for large protein structures.

In the late 1970s, Levitt and Greer first proposed a computational assignment algorithm based on features of angles and distances between residues [21]. Later, Kabsch and Sander proposed another approach, called dictionary of secondary structure of proteins (DSSP) [19]. Given a protein structure, the DSSP method first used the energy functions to calculate the electrostatic interaction energies between carboxyl groups and amino groups of residues, then determined the hydrogen bonds among the structure, and finally identified the α-helices and β-sheets in the protein structure. We note that Carter *et al.* proposed DSSPcont that constitutes a relatively straightforward extension of DSSP by adding continuous assignments [2, 4].

In [28], Richards and Kundrot designed an algorithm called DEFINE that is based on matrix comparisons between inter-atomic distance matrices. Sklenar proposed the P-Curve algorithm which uses curved axes and helicoidal parameters to describe the location of each peptide unit [29]. Basically, as studied by Colloc'h *et al.* [6], the percentage match score between DSSP, DEFINE and P-Curve on a residue per residue basis is around 63%. Maccallum *et al.* determined antiparallel β-sheets based on angles, and used hydrogen bond to decide α-helices and parallel β-sheets [25]. Kleywegt and Jones proposed the YASSPA algorithm, which recognizes α-helices and β-sheets from the distances between C_α atoms [17]. Madsen and Kleywegt developed the DEJAVU algorithm at the level of the fold, which uses the YASSPA algorithm to generate a set of vectors for representing the secondary structure elements [23]. Kleywegt proposed the SPASM algorithm which uses C_α atoms to carry out "fuzzy pattern matching" [16]. Frishman *et al.* and Fodje *et al.* respectively developed the STRIDE and SECSTR algorithms [10, 11]. These two algorithms combine the uses of hydrogen bond energy and statistically derived backbone torsion angle ϕ and ψ information. King and Johnson proposed the XTLSSTR algorithm which uses the angles between C_α atoms and the distances from O_i to N_{i+3} and N_{i+4} [18]. Labesse *et al.* and Martin *et al.* respectively proposed P-SEA and KAKSI that consider the distances between consecutive C_α atoms and the values of ϕ and ψ angles to assign α-helices, β-sheets and coils [20, 24]. Dupuis *et al.* designed VoTAP through Voronoï tessellation that subdivides space into regions with associated discrete points in order to characterize their topological relations [8, 9, 33]. Cubellis *et al.* presented SEGNO, which uses geometric features and ϕ and ψ angles to assign protein structure [5]. Hosseini *et al.* proposed PROSIGN that uses the coordinates of C_α atoms in every window of five consecutive residues to assign protein secondary structures [15].

2 Materials and Methods

Almost all residues of proteins have the same backbone structure. We can imagine each residue in the backbone as a point. Each point has three elements: nitrogen atom, carbon atom and oxygen atom. Then, we can use coordinates of these points to identify α-helices and β-sheets, as described as follows.

Our solution to the assignment problem is divided into two stages. In the first stage, we find out all potential candidates of hydrogen bonds. In the second stage, we would assign α-helix and β-sheet annotations from the output of the first stage. Prior to the assignment process, we determine hydrogen bonds between residues from carboxyl groups and amino groups (Figure 1). Pirard *et al.* have observed three major conditions required for the formation of a hydrogen bond [26].

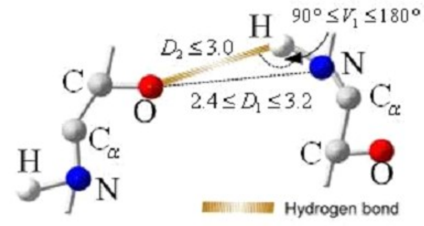


Figure 1: A geometrical description of hydrogen binding.

As shown in Figure 1, these three conditions are (1) D_1 is the distance between an oxygen and a nitrogen that is between 2.4 and 3.2 angstrom, (2) D_2 is a hydrogen bond between a hydrogen and an oxygen whose distance is smaller than or equal to 3.0 angstrom, and (3) V_1 is the angle between NH bond and OH bond whose value is between $90°$ and $180°$.

The difficulty to apply Pirard's conditions to detecting hydrogen bound comes from the fact that the information of D_2 and V_1 are not recorded in the PDB file of a protein structure. Therefore, in the first stage, we use the condition for D_1 to collect all potential candidates of hydrogen bounds. Although this set may contain more false positive instances than Pirard's method, in fact, they are sufficient for us to achieve the task for assigning α-helices and β-sheets in the second stage of our algorithm.

Based upon the above discussion, we formulate a problem, called the *fixed-range nearest neighbor searching problem with constraint*, that is to find all possible candidates of hydrogen bonds in a given protein strucutre. More precisely, let O_i be the coordinate of the oxygen atom in the ith residue and let N_j be the coordinate of the nitrogen atom in the jth residue, where i, j=1,2, $\ldots$,n. The goal of the above problem is to find all pairs (i, j) satisfying two constraints as follows:

1. The distance d between O_i and N_j ranges from 2.4 angstrom to 3.2 angstrom (i.e., $2.4 \leq d \leq 3.2$).

Algorithm 1: A multidimensional divide-and-conquer algorithm to find all possible candidates of hydrogen bonds

Input : A set $S = \{O_1, O_2, \ldots, O_n, N_1, N_2, \ldots, N_n\}$ of $2n$ points in three-dimensional space, d_{min} and d_{max}.

Output: Find all ordered pairs (O_i, N_j) such that the distance from O_i to N_j is between d_{min} and d_{max} for $|i - j| \geq 3$.

Procedure: *3D-Search algorithm*(S')

Step 1 Sort all points in S' according to their y-coordinates.

Step 2 If the number of points in S' is equal to 1, return $\emptyset$.

Step 3 Find a median plane L_{2D} parallel to the (x, z) plane that divides S' into two sets S_{L_3} and S_{R_3} of almost half of points in each set such that the points of S_{L_3} have y-coordinates less than or equal to L_{2D}'s and the points of S_{R_3} have y-coordinates greater than L_{2D}'s. Every point in S_{L_3} lies to the left of L_{2D} and every point in S_{R_3} lies to the right of L_{2D}.

Step 4 Recursively apply ***3D-Search*** on S_{L_3} and S_{R_3}, individually.

Step 5 Project the points of (S') within the slab bounded by $L_{2D} - d_{max}$ and $L_{2D} + d_{max}$ onto the (x, z) plane L_{2D}. Apply ***2D-Search*** on the projected points.

Procedure: *2D-Search algorithm*(S')

Step 1 Sort all points in S' according to their x-coordinates.

Step 2 If the number of points in S' is equal to 1, return $\emptyset$.

Step 3 Find a median line L_{1D} parallel to the x-axis that divides S' into two sets S_{L_2} and S_{R_2} of almost half of points in each set such that the points of S_{L_2} have x-coordinates less than or equal to L_{1D}'s and the points of S_{R_2} have x-coordinates greater than L_{1D}'s. Every point in S_{L_2} lies to the left of L_{1D} and every point in S_{R_2} lies to the right of L_{1D}.

Step 4 Recursively apply ***2D-Search*** on S_{L_2} and S_{R_2}, individually.

Step 5 Find each ordered (O_i, N_j) pair of points such that its distance is between d_{min} and d_{max} for $|i - j| \geq 3$.

2. $|i - j| \geq 3$.

The purpose of our method is to find out all possible candidates for hydrogen binding. If all pairs of carboxyl groups and amino groups are calculated by using exhaustive search, it will take $O\left(n^2\right)$ time. It is still time-consuming and its further improvement is required to solve the original problem efficiently.

Let $d_{min} = 2.4$ and $d_{max} = 3.2$. Algorithm 1 solves the fixed-range nearest neighbor searching problem with constraint problem by using divide-and-conquer approach.

Initially, given a file of the secondary structure of a protein, we extract all oxygen coordinates on carboxyl group and nitrogen coordinates on amino groups from this file as the set of S. Then we feed S to *the 3D-search algorithm*. Finally, we get all possible candidates of hydrogen bonds. The time complexity is $O\left(n \log^2 n + m\right)$ using multidimensional divide-and-conquer algorithm [3], where is the number of identified hydrogen bonds. Potentially, m could be $\Omega\left(n^2\right)$ if all points are within the range. Fortunately, this "cluster state" does not exist in a natural protein. Therefore, m is always dominated by $n \log^2 n$.

In the second stage of our algorithm, α-helix and β-sheet assignments would be considered separately from the output of the first stage. The idea of the second stage is described as follows. We use $\overrightarrow{CO}$ to denote a vector starting from the carbon atom to the oxygen atom in the carboxyl

Table 1: An α-helix of 1MBC

Amino Acid (Position)	Atom	X	Y	Z
T(51)	C	31.018	18.473	-2.102
	O	30.883	19.460	-1.402
E(52)	C	31.589	17.610	0.643
	O	31.305	18.196	1.635
A(53)	C	33.858	19.368	1.235
	O	34.104	19.892	2.403
E(54)	C	32.263	21.927	1.201
	O	32.320	22.945	1.902
M(55)	C	30.299	21.403	3.370
	O	29.934	22.205	4.205
K(56)	C	32.433	21.064	5.573
	O	32.350	21.372	6.774
A(57)	C	33.620	24.117	5.239
	O	34.220	25.011	5.840

group of a residue. Since an α-helix can be viewed as a series of turns that do not occur at the beginning or ending position of the chain, it follows that all of its $\overrightarrow{CO}$ vectors are parallel. Based on this rule, our algorithm calculates the CO-vector for each residue one-by-one and finds all maximal substructures whose CO-vectors are parallel. As for assigning β-sheet, a β-sheet consists of two or more strands. Each strand has the shape of a slab. Our algorithm detects β-sheet from consecutive hydrogen bonds that is non-α-helical.

Next, we discuss how to assign α-helix precisely. In a normal PDB file, each residue on the backbone has a carbon atom connecting to oxygen atom. That is, we can calculate a vector from a carbon atom to an oxygen atom in each residue. This defines a CO-vector in the previous paragraph. For example, Table 1 lists part of coordinates of the protein 1MBC, which can be found from the Protein Data Bank, and it is known as an α-helix.

Since a protein structure can be displayed by a sequence of three-dimensional points, we use RASMOL software to render the structure in Table 1, as shown in Figure 2.

Figure 2 illustrates a sample α-helix. We find that all CO bonds are nearly parallel. Sreerama and Woody proposed that all CO bonds are nearly parallel in an α-helix [32]. Besides, in [19], Kabsch and Sander developed a theory that one α-helix consists of consecutive turns such that a turn can be either a 3-turn, 4-turn or 5-turn. As a result, a helix has at least four nearly parallel each of which is located on one residue in a consecutive chain of

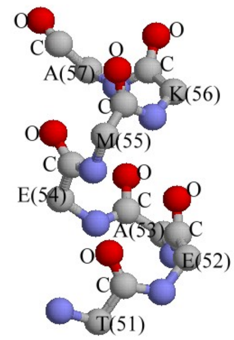


Figure 2: An α-helix of 1MBC.

those residues.

We use $\overrightarrow{CO}$ vectors to determine whether they are nearly parallel between CO bonds. The vector calculation is simple. In Table 1, we leave coordinates of a C atom and an O atom in the first residue. The first C atom is located at $(31.018, 18.473, -2.102)$ and the first O atom is at $(30.883, 19.460, -1.402)$. These two atoms define a vector which is $[30.883 - 31.018, 19.46 - 18.473, -1.402 + 2.102] = [-0.135, 0.987, 0.7]$. In the same way, we can calculate every $\overrightarrow{CO}$ vector for every pairs of a C atom and an O atom.

That is, suppose that a protein structure consists of n residues $a_1a_2\ldots a_n$ where $1 \leq i \leq n$. Then its $\overrightarrow{CO}$ vectors, as defined above, are denoted by $A_1A_2\ldots A_n$ where A_i is a vector defined by the three-dimensional coordinates of between the C atom and the O atom of a_i.

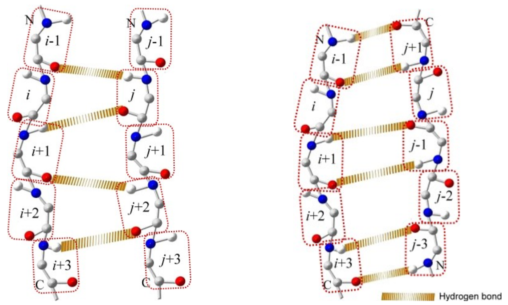
The following formula can be used to determine the angle between two vectors A_i and A_j:

$$\cos\theta\,(i,j) = \frac{A_i \cdot A_j}{\|A_i\|\,\|A_j\|} \quad \text{for } i,j = -1,0,1,2 \qquad (1)$$

where $A_i \cdot A_j$ is the inner product of A_i and A_j, and $\|A_i\|$ is the norm of A_i, similarly for $\|A_j\|$.

Let the protein be characterized by $A_1A_2\ldots A_n$. Now, we consider every four consecutive vectors A_{i-1}, A_i, A_{i+1} and A_{i+2}. They form an α-helix if the angle between each others is rather small. In the other words, an α-helix is formed by four conditions, the angle between A_i and A_{i+1} is small, similarly for A_{i+1} and A_{i+2}, A_{i+2} and A_{i-1}, and A_i and A_{i-1}.

Let $\theta\,(i,j)$ denote the angle between vectors A_i and A_j. We say A_i and A_j are parallel if and only if $\cos\theta\,(i,j) > 0$. When the four consecutive



(a) An parallel ladder (b) An antiparallel ladder

Figure 3: Two types of β-sheet.

vectors satisfy Formula (1), the α-helix is found.

Formula (1) is a complete detecting α-helix equation. We can only consider the numerator in Formula (1) because the denominator is always larger than zero. The simplified formula is shown in following:

$$D\left(i,j\right) = A_i \cdot A_j \qquad i,j \in \{-1,0,1,2\} \tag{2}$$

where $D\left(i,j\right)$ is a detecting score. Here, the significance of $D\left(i,j\right)$ is as same as $\cos\theta\left(i,j\right)$.

Based upon the above concepts, the length of an α-helix can stretch more than four residues. Suppose that $A_{i-1}A_iA_{i+1}A_{i+2}$ is an α-helix, we calculate a new angle between A_{i+3} and A_{i+4}, and so on. The extension process in next residue is stopped when the accumulated angle is smaller than 0.

Given a protein structure, we can find all α-helices when structure was scanned completely at a time. Using our algorithm, the time complexity of detecting α-helix is $O\left(n\right)$, where n is the number of residue.

Inside a β-sheet, there are again hydrogen bonds such that these hydrogen bonds are perpendicular to the backbone. Each strand has the shape of a slab.

Following Kabsch's definition, we let predicate $Hbond(i,j)$ denote a hydrogen bond between the oxygen atom of the ith residue and the hydrogen atom of the jth residue. A β-sheet comprises bridge, β-ladder and β-sheet [19]. Then the predicate for a parallel $bridge\left(i,j\right)$ is defined as $\left[Hbond\left(i-1,j\right)\text{ and } Hbond\left(j,i+1\right)\right]$ or $\left[Hbond\left(j-1,i\right)\text{ and } Hbond\left(i,j+1\right)\right]$, and the predicate for an antiparallel $birdge\left(i,j\right)$ is defined as $\left[Hbond\left(i,j\right)\text{ and } Hbond\left(j,i\right)\right]$ or $\left[Hbond\left(i-1,j+1\right)\text{ and } Hbond\left(j-1,i+1\right)\right]$. A β-ladder is a set of one or more consecutive bridges of identical type, and a β-sheet is a set of one or more β-ladders connected by shared residues.

Figure 3 illustrates two types of β-sheets. That is, a parallel ladder and an antiparallel ladder. In the parallel ladder as shown in Figure 3(a), $i-1$ is paired with j, similarly for pairs between j and $i+1$, and $i+1$ and $j+2$. In Figure 3(b), the antiparallel ladder consists of consecutive pairs of resides, $i-1$ and $j+1$, $i+1$ and $j-1$, and $i+3$ and $j-3$.

The β-sheet assignment is a difficult task because a β-sheet may consist of more than one strand such that these strands have cluster property in three-dimensional space. We must filter out α-helix from all residues of hydrogen bond. After filtering out those α-helices, we can easily identify the candidates of β-sheet from the remaining residues with hydrogen bonds.

Let $PHbond\left(i,j\right)$ be the predicate representing that there is an identified hydrogen bond between O_i and N_j. Then, according to the definition of Kabsch, we can detect parallel or antiparallel bridges by inspecting whether $PHbond(j, i+k)$ is true for all $k = -2, 0, -2$. However, in implementation, we also check $k = -1, 1$ for fault tolerance because that $Hbond\left(j,i-1\right)$ holds may be caused by an inaccurate identification of coordinates of oxygen and nitrogen atoms from the actual case that $Hbond\left(j,i\right)$ or $Hbond\left(j,i-2\right)$ is true.

The time complexity for the second stage of our method depends on the total time required for the assignments of α-helices and β-sheets, which is equal to $O\left(m\right)$. The overall time complexity of our algorithm is thus $O\left(n\log^2 n\right) + O\left(m\right) = O\left(n\log^2 n + m\right)$.

3 Experiments

In the following, we show some experiments using our assignment algorithm, as well as six other methods (DSSP, DEFINE, STRIDE, P-SEA, DSSPcont and Voro3D), for assigning the secondary structures in some proteins.

- **Experiment 1**: A Protein 1EW4

  Protein 1EW4 is a CyaY protein of Escherichia coli. For illustrative purpose, we input complete coordinates of 1EW4 into our algorithm and six other assignment methods. The results are shown in Figure 4.

  As can be seen in Figure 4, the secondary structures of the Protein CyaY is well assigned by each method.

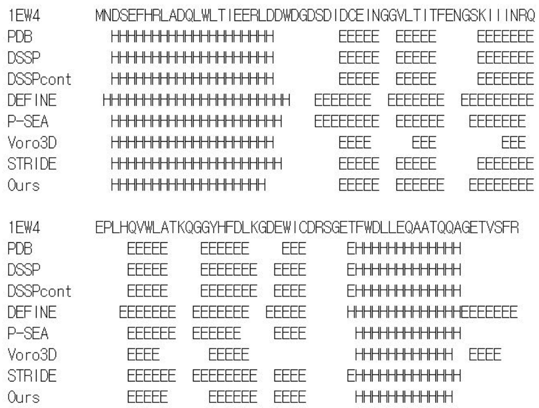


Figure 4: Comparison of assignments done by PDB, DSSP, DSSPcont, DEFINE, P-SEA, Voro3D, STRIDE and our method. H represents the α-helix class and E represents β-sheet class.

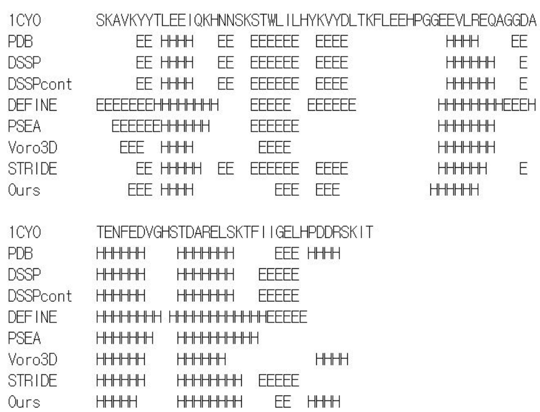


Figure 5: The experimental results for a difficult case.

- **Experiment 2**: A difficult assignment instance: Protein 1CYO Dupuis *et al.* proposed a protein 1CYO whose secondary structures are hard to be assigned correctly by currently available method [9]. We use this protein as the input data for this experiment and its results are shown in Figure 5.

Our experimental results show that all the methods have similar assignment results. Particularly note that DSSP, DSSPcont and STRIDE have the same result.

Finally, we use a set of 194 proteins which was selected by Dupuis [9] from the CheckBank to further test our tool and also compare our results with those obtained by DSSP, VoTAP, STRIDE, P-SEA, DSSPcont, STRIDE, DEFINE and CONSENSUS.

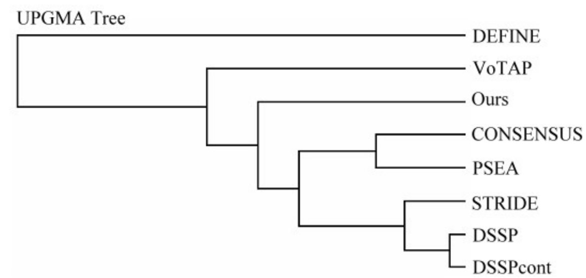


Figure 6: The phylogenetic tree for the eight methods in Table 2 obtained by the UPGMA tree-construction algorithm.

We use two parameters defined below to measure the performance of all methods: Given the assignment results of two given methods, AP denotes the number of residues that are assigned as the same type of secondary structures by both methods and AN the number residues that are assigned as different types of secondary structures. Let agreement A be the ratio of the number of residues with which both methods agree to the total number of residues. More precisely, we have

$$A = \frac{AP + AN}{\text{the number of residues}} \tag{3}$$

The result comparisons for each pair of the above assignment methods based on the above measure are shown in Table 2.

In Figure 6, we use the UPGMA algorithm to display the experimental results of Table 2 [31]. We found that the DEFINE algorithm is very different from the other algorithms. Generally speaking, the experimental results show that our algorithm has competitive result as the other six methods, particularly including the DSSP that is currently the most widely used tool.

4 Conclusion

In this paper, we proposed a simple method for effectively solving the protein structure assignment problem. In this study, we first identified all possible candidates of hydrogen bonds from the Euclidean distance between oxygen and nitrogen atoms on the backbone of a protein. We then applied the multidimensional divide-and-conquer algorithm to determine pairs of residues that contain at least one hydrogen bond. The time-complexity of our algorithm is $O\left(n \log^2 n + m\right)$, where n is the number of residues and m is the number of

Table 2: The pairwise agreement in secondary structure assignment given by eight methods

	PSEA	DSSP	DSSPcont	CONSENSUS	VoTAP	DEFINE	Ours
STRIDE	88.49%	94.76%	95.81%	90.36%	90.30%	68.37%	86.32%
PSEA		83.02%	83.79%	93.10%	76.57%	69.52%	80.12%
DSSP			98.81%	90.08%	79.16%	64.59%	87.11%
DSSPcont				90.32%	79.38%	66.82%	86.70%
CONSENSUS					74.35%	74.19%	79.36%
VoTAP						53.42%	79.86%
DEFINE							59.52%

identified hydrogen bonds. In α-helix assignment, we considered a protein as a sequence of $\overrightarrow{CO}$ vectors. Then we determined whether nearby $\overrightarrow{CO}$ vectors are parallel. When at least four consecutive parallel $\overrightarrow{CO}$ vectors were found, we may say that their corresponding residues were considered as a part of an α-helix. In β-sheet assignment, our algorithm identified specific patterns of β-sheets on the linking of hydrogen bonds. To demonstrate its effectiveness, we used a set of 194 proteins from the CheckBank to test our tool that was implemented based on our algorithm. The experimental results revealed a fact that seven out of the eight assignment methods, including our method, exhibit strikingly agreements among one another. Therefore, we conclude that our algorithm can effectively and quickly solve the protein secondary structure assignment problem.

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